Joseph Samuel Canzano

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EDUCATION

Ph.D.	**University of California, Santa Barbara** GPA: 3.81	**Dynamical Neuroscience**	2026
M.S.	**University of Florida** Concentration: Neuroscience GPA: 3.95	**Medical Science**	2019
B.S.	**University of North Florida** Concentration: Molecular/Cell Biology & Biotechnology Minor: Chemistry	**Biology**	2015

EMPLOYMENT & RESEARCH EXPERIENCE

2019-	**Graduate Student Researcher** PI: Spencer Smith	UCSB ECE Dept.
2017-2019	**Graduate Research Asst.** PI: Dr. Karim Oweiss	University of Florida ECE Dept.
2016-2017	**Biological Scientist I** PI: Dr. Martha Campbell-Thompson	University of Florida Experimental Pathology
2013-2015	**Undergraduate Research Asst.** PI: Dr. Evette Radisky	Mayo Clinic Cancer Biology
2011-2012	**Undergraduate Research Asst.** PI: Dr. John Hatle	University of North Florida Biology Dept.

PUBLICATIONS

In prep Canzano J.S., Rader J., Duron M., Smith S.L., “Mesoscale population coding of action and vision supporting visual navigation in complex, open-world VR”

2026 Rajpal, H., Stefens, C., Saeedian, M., **Canzano, J.S.**, Kareithi, M.G., Barahona, M., Smith, S.L., Schultz, S.R., Jensen, H.J. “Synergy mediates

Long-Range Correlations in the Visual Cortex Near Criticality" **Front. Comput. Neurosci.**

2025 *Schneider, M., ***Canzano, J.S.**, *Peng, J., *Hou, Y., Smith, S.L., Beyeler, M. "Mouse vs. ai: A neuroethological benchmark for visual robustness and neural alignment" **ArXiv, *equal contributions**

2025 Yu, C.H., Yu, Y., **Canzano, J.S.**, Fei, Y., Smith, S.L. "Non-inertial scan angle multiplier for expanded fields-of-view" **Biorxiv**

2023 Antoniades, A., Yu, Y., **Canzano, J.S.**, Wang, W., Smith, S.L. "Neuroformer: Multimodal and Multitask Generative Pretraining for Brain Data" **Arxiv**

2020 **Canzano J.S.**, Subramanian N, Castro R, Siddiqi A, Oweiss K, "In vivo two-photon imaging and parasympathetic neuromodulation of pancreatic microvascular dynamics in rats" **Biorxiv**

2018 **Canzano, J.,** Nasif L.H., Butterworth E.A., Fu, A., Atkinson, M., Campbell-Thompson, M. "Islet Microvasculature Alterations with Loss of Beta-Cells in Patients with Type 1 Diabetes." **J Histochem Cytochem** [PMID: 29771178]

2014 Tokar, D.R., Veleta, K.A., **Canzano J.**, Hahn, D.A., Hatle, J.D. "Vitellogenin RNAi Halts Ovarian Growth and Diverts Reproductive Proteins and Lipids in Young Grasshoppers." **Integr. Comp. Biol.** [PMID: 24920749]

TALKS/PRESENTATIONS

2026 Poster — V Lin, JS Canzano, SL Smith. "Behavioral sequence analysis of mouse visual navigation in virtual reality" UCSB URCA Colloquium. Undergraduate mentee poster.

2025 Workshop — *M Schneider, ***JS Canzano**, *J Peng, *Y Hou, SL Smith, M Beyeler "Mouse vs. AI: A Neuroethological Benchmark for Visual Robustness" **NeurIPS 2025**

2025 Poster — H Rajpal, C Stefens, M Saeedian, **JS Canzano**, M Barahona, S Schultz, SL Smith, HJ Jensen, "Synergistic information processing as the mechanism for long-range correlations in the brain." **Network Neuroscience 2025**

2024 **Invited Talk** — **JS Canzano**, M Duron, SL Smith, "Large scale population coding during open-world behavior guided by complex visual stimuli" **Institute of Science, Tokyo.** Host: Riichiro Hira

2024	**Talk**	**JS Canzano**, M Duron, SL Smith, “Large scale population coding during open-world behavior guided by complex visual stimuli” **Japan Neuroscience Society Conference 2024**
2024	Poster	C Stefens, H Rajpal, **JS Canzano**, M Saeedian, M Yang, LD Arcangelis, HJ Jensen, M Barahona, SL Smith, SR Schultz, “Mesoscopic multiphoton calcium imaging reveals a confluence of overlapping avalanches with varying distance to criticality and distinct roles” **SfN 2024**
2024	Poster	A Antoniades, Y Yu, **JS Canzano**, WY Wang, SL Smith, “Neuroformer: Multimodal and Multitask Generative Pretraining for Brain Data” **ICLR 2024**
2023	Poster	**JS Canzano**, M Duron, SL Smith, “Navigation and high-level visual features in mouse posterior cortex”, UCSB ECE Graduate Summit
2019	Poster	N. SUBRAMANIAN, ***JS CANZANO**, B. GOOLSBY, H. KIM, P. NAVARRO, R. CASTRO, K. G. OWEISS “Simultaneous two-photon calcium imaging and cell-targeted optogenetic stimulation shapes somatosensory cortex plasticity” **SfN 2019, *first author** placed second for registration technicality
2019	Poster	B GOOLSBY, HJ KIM, **JS CANZANO**, K. G. OWEISS “All-optical functional mapping of association cortex subserving working memory and perceptual decision making” **SfN 2019**
2019	**Talk**, Poster	**JS Canzano**, N Subramanian, A Siddiqi, R Castro, R Johnson, K Oweiss, “In Vivo Two-Photon Imaging of Pancreatic Sympathetic and Parasympathetic Microvessel Neuromodulation” **11th Congress of the International Society for Autonomic Neuroscience**
2019	Poster	**JS Canzano**, H Kim, R Castro, K Oweiss “Artificial Forelimb Somatosensation via Optogenetic Stimulation and Two-Photon Calcium Imaging in a Sensory-Guided Motor Task” **Sculpted Light in the Brain Conference**
2019	Poster	**JS Canzano**, R Castro, H Kim, K Oweiss, “Artificial Forelimb Somatosensation via Optogenetic Stimulation and Two-Photon Calcium Imaging in a Sensory-Guided Motor Task”, **Max Planck Florida Institute Sunposium 2019**

2017	**Talk**, Poster	"Endocrine and Exocrine Microvascular Alterations in Human Type 1 Diabetes," **JDRF nPOD Annual Scientific Meeting**
2016	Presentation	"Endocrine and Exocrine Microvascular Alterations in Human Type 1 Diabetes," **UF Diabetes Institute "Works in Progress"**
2013	Presentation	"Understanding mesotrypsin: The role of residue 193 in the unusual reverse activity of human mesotrypsin and similar vertebrate enzymes," **UNF Mayo Clinic Intern Talks**
2012	Poster	"Vitellogenin RNAi halts ovarian growth and diverts reproductive proteins and lipids in young grasshoppers," **UNF SOARS Conference**
2012	Poster	"Physiological effects of RNAi for vitellogenin on reproduction and fat body storage in grasshoppers," **Society for Integrative Comparative Biology Annual Meeting**

TEACHING

TA positions:

- **PSY 111L Behavioral Neuroscience Lab**, UCSB. Instructor: Ron Keiflin
- **PSY 115 Neuropharmacology**, UCSB, Instructor: Karen Szumlinski
- **PSY 106 Intro Biopsych,** UCSB. Instructor: Samantha Scudder
- **PSY 111 Behavioral Neuroscience**, UCSB. Instructor: Vanessa Woods
- Reader, **PSY161 Systems Neuroscience**, UCSB. Instructor: Michael Goard

Training:

- UCSB PBS TA training course series

MENTORSHIP

Undergraduate mentees:

- Vivian Lin, quantitative behavior analysis, 2025-present
- Janelle Rader, animal behavior, 2022-2025
- Miranda Duron, animal behavior, 2020-2022
- Abdurahman Siddiqi, peripheral neuromodulation, 2018-2019
- Lith Nasif, peripheral neural immunofluorescence, 2016-2017

SELECTED ACADEMIC COURSEWORK

Mo/Cell Bio & Neurobio   Systems, Dynamics, Machine Learning   Chemistry & Physics

Graduate	Undergraduate
Computational Neuroscience (UCSB)	Molecular/Cell Biology
Machine Learning (UCSB)	Genetics
Linear Systems (UCSB)	Biochemistry
Systems Neuroscience (UCSB)	Physiology
ODE's (dynamical systems) (UCSB)	Virology
Biological dynamics (UCSB)	Computer Science
Measuring/Manipulating Neural Activity (UCSB)	Multivariate Calculus
Neuropharmacology (UF)	Physical Chemistry
Molecular/Cell Neuroscience (UF)	Inorganic Chemistry
Neural Signals, Systems & Tech. (UF)	Psychobiology
Functional Human Neuroanatomy (UF)	Statistics
Fund. of Biomedical Science (UF)	

WORKSHOP PARTICIPATION

2022 **Neuropixels and OpenScope Workshop**, Allen Institute, **2022**

PROFESSIONAL MEMBERSHIPS

2024- Japanese Neuroscience Society
2019- Society for Neuroscience

AWARDS AND HONORS

2019 **Travel award, Sculpted Light in the Brain 2019**
2013 Awarded joint UNF Biology - Mayo Clinic internship

REFERENCES

Available on request